

NODEWISE ERROR CERTIFICATES FOR RECURSIVELY PROJECTED N -ARY COMPOSITION TREES

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ABSTRACT. We study the worst-case discrepancy between ambient evaluation and recursive orthogonal projection on a finite typed tree of multilinear laws, with local law norm at most M , closure residual at most ρ , and k internal vertices. An exact child-error subset expansion yields the universal bounds $E^{\text{amb}} \leq k\rho M^{k-1}L$ and $E^P = E^{\text{red}} \leq (k-1)\rho M^{k-1}L$, together with mixed-block dynamic-programming and residual path-sum certificates. A pair-exchange argument gives the globally optimal slot order within the declared scalar telescoping family. A gated planar-rotation construction matches the ambient coefficient in registered cells, but does not establish fixed-positive- $\eta = \rho/M$ sharpness of the projected coefficient $k-1$. Directed interval calculations certify 60 small cells globally; across the full gap registry the maximum unresolved absolute and relative gaps are 32 and 1. The finite base study contains 15,493 scientific instances, 81,445 enumerated tree occurrences, and 1,530 exhaustive leakage masks, with CPU/GPU difference at most 1.922×10^{-8} . The main limitations are that theorem-level novelty is not established, the extended adversarial grid is resource-gated, and most fixed- η constants remain open; the release status is therefore `FAIL_CLOSED_NOVELTY`.

1. INTRODUCTION

Finite compositions of multilinear operations occur in nonassociative identities, colored-operad syntax, hierarchical tensor calculations, and structure-preserving reduction. A common computational operation is to project the output of every internal node back to a selected subspace. Even when every local projection defect is small, nested composition can amplify or cancel those defects in a topology-dependent way. A global norm product is safe but can obscure the source, direction, and interaction order of the error.

This paper fixes a finite typed ordered tree and asks a precise optimization question: among laws, compatible orthogonal projectors, and unit reduced leaf data satisfying declared local norm and closure constraints, what is the best constant controlling ambient, projected-root, normal-root, or reduced-coordinate error? We separate four logically distinct tasks: proving universal upper bounds, constructing admissible lower examples, certifying small global optima, and searching larger instances empirically. The last task is never used as a proof.

The strongest universal result is the difference between k and $k-1$. The ambient and normal errors may include a new residual at the root, whereas the projected and reduced root errors cannot. This removes exactly one source in a homogeneous tree. The coefficient is a theorem-level upper bound; its fixed- η optimality is not inferred from the form of the proof.

2. CONTRIBUTIONS AND EPISTEMIC STATUS

The contributions are mapped one-to-one to registry entries and evidence:

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- (1) an exact local subset expansion and exact orthogonal relationship among the four named root errors (Theorem 6.1 and Proposition 5.1);
- (2) universal k and $k - 1$ homogeneous coefficients valid for heterogeneous types, laws, and arities under uniform M, ρ (Theorem 9.1);
- (3) a mixed-state dynamic program, residual path sums, and a globally optimal ordering theorem within the scalar telescoping family (Theorems 7.1 and 8.1);
- (4) certified explicit lower constructions, exact small symbolic cases, and independent empirical searches with explicit optimality gaps; and
- (5) a deduplicated finite experiment and artifact protocol that tests sound bounds, topology effects, signed forests, CP/projection separation, precision, and CPU/GPU parity.

The exact expansion and ambient coefficient are conservative specializations of standard multilinear analysis. The projected-root, mixed-mask, and path-sum formulations were not located verbatim in the bounded prior-art search, but absence from that search is not novelty evidence. Every candidate novelty claim therefore remains NOVELTY_NOT_ESTABLISHED and every theorem remains PENDING_HUMAN_REVIEW.

3. PRIOR ART

Typed tree composition is standard in colored operads [15, 12]. Filippov’s (n)-Lie identity and the Akivis–Sabinin landscape motivate signed nonassociative forests [4, 14]. Approximate multiplicativity in Banach algebras studies stability of maps under different global hypotheses [8]; it is adjacent to, but does not directly solve, the fixed-tree recursive projection problem.

Forward and backward error analysis already make composition order and local perturbation propagation central [7]. Layered Lipschitz certificates and sound abstract interpretation likewise demonstrate that architecture-aware summaries can improve naive products [2, 5]. These are strong prior-art reasons not to claim that a dynamic program is intrinsically new.

Hierarchical tensor formats use dimension trees and topology-adaptive approximation [6, 1]. Their principal error is representation or rank truncation, while the present principal error is a normal closure defect created at every algebraic node. CP approximation, factor nonuniqueness, and identifiability are kept separate [9, 10]. Projection-based structure-preserving model reduction supplies further adjacent precedent [3].

Validated interval arithmetic and moment/SOS methods provide global certification tools when their hypotheses and solver statuses are verified [13, 11]. V3 uses directed interval recursion and exact symbolic elimination in selected low-dimensional cases; it does not label an imported SDP interface as a certificate merely because the package imports.

4. TYPED COMPOSITION TREES

Let $\mathfrak{T}$ be a finite set of types. For each $\tau \in \mathfrak{T}$, let V_τ and W_τ be finite-dimensional real or complex Hilbert spaces, let $Q_\tau : W_\tau \rightarrow V_\tau$ be an isometry, and put $P_\tau = Q_\tau Q_\tau^*$. An internal vertex v of arity a_v carries a bounded law

$$\mu_v : \prod_{j=1}^{a_v} V_{\tau(v,j)} \longrightarrow V_{\tau(v)}.$$

Each ordered child edge must match its declared input type. Type-invalid trees have no numerical semantics and are rejected before contraction.

5. AMBIENT AND RECURSIVELY PROJECTED EVALUATIONS

For reduced leaf data $z_\ell \in W_{\tau(\ell)}$, define

$$F_\ell = R_\ell = Q_{\tau(\ell)} z_\ell, \quad F_v = \mu_v(F_{c_1}, \dots, F_{c_{a_v}}), \quad R_v = P_{\tau(v)} \mu_v(R_{c_1}, \dots, R_{c_{a_v}}).$$

At the root, write $F = F_T$, $R = R_T$, and $P = P_{\tau(T)}$. The four errors are

$$E_T^{\text{amb}} = \|F - R\|, \quad E_T^P = \|PF - R\|, \quad E_T^N = \|(I - P)F\|, \quad E_T^{\text{red}} = \|Q^*F - Q^*R\|.$$

Proposition 5.1 (Exact root geometry). *For every valid typed tree,*

$$(E_T^{\text{amb}})^2 = (E_T^P)^2 + (E_T^N)^2, \quad E_T^{\text{red}} = E_T^P.$$

The proof is the orthogonal decomposition $F - R = (PF - R) + (I - P)F$; Appendix A records the complete argument. Orthogonality is essential and is tested independently in the reference and vectorized evaluators.

6. EXACT LOCAL ERROR DECOMPOSITION

At a node v , abbreviate $F_i = F_{c_i}$, $R_i = R_{c_i}$, and $D_i = F_i - R_i$. Define

$$r_v = (I - P_v) \mu_v(P_{c_1} \cdot, \dots, P_{c_a} \cdot).$$

Theorem 6.1 (Exact child-error subset expansion). *For $y_i^S = D_i$ if $i \in S$ and $y_i^S = R_i$ otherwise,*

$$D_v = r_v(R_1, \dots, R_a) + \sum_{\emptyset \neq S \subseteq [a]} \mu_v(y_1^S, \dots, y_a^S).$$

The term indexed by S records exactly which child branches carry an error. Singleton subsets are first-order propagated errors; larger subsets are genuine branch interactions. Applying P_v removes the local residual exactly but not the propagated interactions.

7. DYNAMIC-PROGRAMMING CERTIFICATES

For each node, the implementation stores $B_v^F, B_v^R, B_v^A, B_v^P, B_v^N$ and certified mixed-block norms. Splitting each child error as $D_i = D_i^P + D_i^N$ creates three slot states: recursively projected R , projected error D^P , and normal error D^N . Every state vector other than all R is bounded separately.

Theorem 7.1 (Sound mixed-mask recursion). *The three-state recursion returns valid full, projected, and normal root certificates. Its complexity is*

$$O(|T|3^{a_{\max}} + |T|a_{\max} \log a_{\max}),$$

and is linear in $|T|$ for the declared arities two through four.

8. OPTIMAL TELESCOPING ORDER

For a single multilinear replacement, let e_i bound the slot error, r_i, f_i the projected and ambient child magnitudes, and G_i a slot-specific gain. Set $w_i = G_i e_i$ and

$$C(\pi) = \sum_t w_{\pi_t} \prod_{s < t} r_{\pi_s} \prod_{s > t} f_{\pi_s}.$$

Theorem 8.1 (Globally optimal order within the scalar family). *Put $d_i = f_i - r_i$. Sort positive d_i , then zero d_i , then negative d_i , and within each nonzero sign class sort w_i/d_i increasingly. The resulting permutation globally minimizes $C(\pi)$.*

The adjacent exchange condition is $w_i(f_j - r_j) \leq w_j(f_i - r_i)$. Exhaustive property tests compare the sorting rule against every permutation through arity seven, including zero and negative denominators. The theorem optimizes this scalar telescoping family only.

9. HOMOGENEOUS UNIVERSAL BOUNDS

Let $k(T)$ be the number of internal vertices and $L_T = \prod_\ell \|z_\ell\|$. Assume every node law has norm at most M and every closure map has norm at most ρ .

Theorem 9.1 (Universal coefficients). *For every finite valid typed tree,*

$$E_T^{\text{amb}} \leq k\rho M^{k-1}L_T, \quad E_T^N \leq k\rho M^{k-1}L_T,$$

and

$$E_T^P = E_T^{\text{red}} \leq (k-1)\rho M^{k-1}L_T.$$

The proof is an induction on the tree. Propagated child errors contribute $\sum_i k_{c_i} = k-1$, and the new root residual contributes one additional unit only to ambient and normal error.

10. PROJECTED-ROOT IMPROVEMENT

The coefficient $k-1$ is not a heuristic correction. The root local residual belongs to the normal space and is annihilated by P . In particular, a one-node tree has $E_T^P = E_T^{\text{red}} = 0$, even when its ambient error is nonzero. For a five-input ternary associator represented as two signed two-node trees, the immediate triangle coefficient is therefore $1+1=2$, not the ambient $2+2=4$.

This proof establishes a universal upper. It does not establish that $k-1$ is the best coefficient at every fixed $\eta = \rho/M > 0$, dimension, rank, topology, or repeated-law class.

11. NODEWISE PATH-SUM BOUNDS

Fix a valid telescoping order at each ancestor. If a source residual at node v has magnitude bound $\lambda_v = \rho_v \prod_i B_{c_i}^R$, and $h_{a,j}$ is the certified gain for propagating child j through ancestor a , then

$$B_{T,\text{path}}^{\text{amb}} = \sum_{v \in \text{Int } T} \lambda_v \prod_{(a,j) \in \text{path}(v,\text{root})} h_{a,j}.$$

For projected-root error, the root source is absent and the final edge uses a projected-output gain. The artifact schema records every source contribution so that the total can be independently reconstructed.

12. SHARPNESS CONSTRUCTIONS

The principal explicit family acts on a two-dimensional tangent/normal plane. For $0 \leq \eta \leq 1$, let

$$T_\eta = \begin{bmatrix} \sqrt{1-\eta^2} & -\eta \\ \eta & \sqrt{1-\eta^2} \end{bmatrix}.$$

The first slot is rotated by T_η , while all remaining slots gate on their first coordinate. The multilinear operator norm is one and the local closure norm is exactly η . Embedding this plane into larger ambient dimensions preserves the lower value. Directed interval recursion encloses the resulting normalized tree error.

13. SIGNED POLYNOMIAL EXPRESSIONS

A compatible signed forest is $\mathcal{F} = \sum_\alpha c_\alpha T_\alpha$. The unconditional certificate is the coefficient-weighted sum of per-tree bounds. V3 first combines syntactically identical terms exactly, represents shared subtrees as an expression DAG, and then evaluates the whole signed expression during adversarial search. Only the first operation is an unconditional symbolic cancellation theorem; adversarial values are lower bounds.

14. CERTIFIED SMALL-CASE OPTIMIZATION

An explicit attained value is a rigorous lower bound only after its admissibility is checked. Directed interval arithmetic encloses the gated-rotation recursion. For selected univariate quadratic reductions, an independent exact symbolic routine enumerates the endpoints and stationary points on $[-1, 1]$ and retains the exact nonnegativity witness. An SDP package import is recorded only as availability; no solver status is promoted to a certificate without validation.

15. ADVERSARIAL NUMERICAL SEARCH

The empirical interface provides two independent optimizer families: automatic-differentiation search with Adam followed by L-BFGS, and derivative-free differential evolution. Inputs and tensors are normalized after each admissibility update. Every trajectory, failed status, and independent disagreement is retained. Optimizers can improve a lower bound; they cannot certify that no better point exists.

The pilot run used two seeds and two gradient restarts plus an independent derivative-free recheck. The requested extended schedule contains 460,800 trajectories and remains resource-gated; its absence is a release blocker rather than an omitted detail.

16. GRANULAR EXPERIMENTAL RESULTS

The base matrix contains blocks A–I: exact small cases, universal constant profiles, dimension/rank embeddings, nodewise certificate comparisons, typed controls, exhaustive leakage masks, signed forests, CP/projection budgets, and precision/device checks. Repeated executions are not independent scientific instances. The complete base has 15,493 unique cells; optimizer seeds and restarts that were not run are labeled pending rather than silently collapsed.

17. COUNTEREXAMPLES AND NEGATIVE CONTROLS

The assumptions are operational, not decorative.

- (1) If P is oblique, Proposition 5.1 can fail. For $P = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$ and $D = (0, 1)^\top$, the components $PD = (1, 0)^\top$ and $(I - P)D = (-1, 1)^\top$ are not orthogonal.
- (2) A type-invalid edge cannot be repaired by numerical broadcasting; all 48 declared invalid block-E controls are rejected before evaluation.
- (3) A one-node projected constant is exactly zero, disproving any formula that blindly assigns one local residual to projected-root error.
- (4) Cancellation in a signed forest can make a triangle certificate strict; an empirical lower never reverses the upper-bound implication.
- (5) Larger dimension contains the dimension-two construction, but this embedding alone does not show that additional coordinates improve the optimum.

18. APPROXIMATION PLUS PROJECTION

Let $\hat{\mu}_v$ approximate μ_v with $\delta_v = \|\mu_v - \hat{\mu}_v\|_{\text{op}}$. Inserting the ambient $\hat{\mu}$ -tree separates representation from recursive projection:

$$\|F_\mu - R_{\hat{\mu}}\| \leq \|F_\mu - F_{\hat{\mu}}\| + \|F_{\hat{\mu}} - R_{\hat{\mu}}\|.$$

In the homogeneous k -node case, the transparent budget is

$$\begin{aligned} E_{\text{repr}} &\leq k\delta(M + \delta)^{k-1}L, \\ E_{\text{closure, base}} &\leq c\rho M^{k-1}L, \end{aligned}$$

$$E_{\text{interaction}} \leq c\rho[(M + \delta)^{k-1} - M^{k-1}]L,$$

with $c = k$ for ambient error and $c = k - 1$ for projected-root error. CP reconstruction error is not folded into ρ .

19. LIMITATIONS

The principal limitations are mathematical rather than cosmetic.

- (1) The fixed- η optimal constants are unknown for most projected, typed, repeated-law, and signed-forest families.
- (2) The exact $k - 1$ formulation and mixed-mask specialization have not received an independent expert novelty review.
- (3) Interval enclosures certify explicit constructions, not the global supremum unless an independent upper closes the gap.
- (4) The base matrix retains every requested scientific axis, but the full seed/restart optimizer grid is pending under an explicit resource gate.
- (5) Operator norms are exact only for declared special constructions; general dense and CP laws may use Frobenius upper bounds.
- (6) All results are finite-dimensional. No continuum, infinite-tree, measurability, or discretization-limit theorem is claimed.

20. OPEN PROBLEMS

The central open questions are:

- (1) determine the exact fixed- η projected constant $C_T^P(\eta)$ for nontrivial topology families;
- (2) decide whether dimension or projector rank can improve upon embedded planar constructions;
- (3) obtain cancellation-aware certified constants for associator, FI, GJI, and Jacobiator forests;
- (4) characterize when mixed-mask, path-sum, and optimized telescoping certificates dominate one another;
- (5) replace Frobenius operator bounds by validated tensor spectral-norm certificates at useful scale; and
- (6) complete an independent prior-art and proof audit sufficient to change the novelty status, if justified.

21. CONCLUSION

Recursive orthogonal projection on a typed multilinear tree has an exact local error algebra. That algebra yields a sound source-resolved dynamic program, a closed-form optimal ordering within the scalar telescoping family, and the universal distinction between k ambient sources and $k - 1$ projected sources. The finite evidence is broad enough to expose topology, interaction, precision, and representation effects, but not broad enough to resolve most fixed- η optima or theorem-level novelty. The appropriate outcome is therefore a reproducible research draft with a fail-closed release gate.

APPENDIX A. FULL PROOF APPENDICES

This appendix records the complete finite-dimensional arguments used by the claim registry. No statement below depends on an optimizer, a conjecture, or a numerical plot.

A.1. Magnitude induction and exact root geometry. For a subtree rooted at v , let k_v be its number of internal vertices and let L_v be the product of the reduced leaf norms below it.

Lemma A.1 (Subtree magnitudes). *If every node law has multilinear operator norm at most M , then*

$$\|F_v\| \leq M^{k_v} L_v, \quad \|R_v\| \leq M^{k_v} L_v.$$

Proof. At a leaf, $F_v = R_v = Q_v z_v$, and Q_v is an isometry. Suppose the claim holds for the children $c_1, \dots, c_a$. Multilinearity and the definition of the operator norm give

$$\|F_v\| \leq M \prod_i \|F_{c_i}\| \leq M^{1+\sum_i k_{c_i}} \prod_i L_{c_i} = M^{k_v} L_v.$$

The same argument applies to $R_v = P_v \mu_v(R_{c_1}, \dots, R_{c_a})$, because an orthogonal projector is contractive. Induction proves both claims. $\square$

Proposition A.2 (Root-error orthogonality). *For every valid typed tree,*

$$(E_T^{\text{amb}})^2 = (E_T^P)^2 + (E_T^N)^2, \quad E_T^{\text{red}} = E_T^P.$$

Proof. At the root, write $P = QQ^*$, $F = F_T$, and $R = R_T \in \text{ran } P$. Then

$$F - R = (PF - R) + (I - P)F.$$

The first summand belongs to $\text{ran } P$ and the second to its orthogonal complement. Pythagoras gives the first identity. Moreover,

$$Q^*F - Q^*R = Q^*(PF - R).$$

The restriction of Q^* to $\text{ran } P$ is an isometry, which proves the second identity. $\square$

A.2. Exact subset expansion.

Theorem A.3 (Local child-error expansion). *Let v have arity a , children c_i , recursively projected values R_i , and errors $D_i = F_i - R_i$. For*

$$y_i^S = \begin{cases} D_i, & i \in S, \\ R_i, & i \notin S, \end{cases} \quad r_v = (I - P_v) \mu_v(P_{c_1} \cdot, \dots, P_{c_a} \cdot),$$

one has the exact identity

$$D_v = r_v(R_1, \dots, R_a) + \sum_{\emptyset \neq S \subseteq [a]} \mu_v(y_1^S, \dots, y_a^S).$$

Proof. Since $F_i = R_i + D_i$, repeated multilinearity yields

$$\mu_v(F_1, \dots, F_a) = \sum_{S \subseteq [a]} \mu_v(y_1^S, \dots, y_a^S).$$

The term indexed by the empty set is $\mu_v(R_1, \dots, R_a)$, while $R_v = P_v \mu_v(R_1, \dots, R_a)$. Subtracting R_v separates the empty-set term into its normal component $r_v(R_1, \dots, R_a)$; all nonempty terms remain unchanged. No inequality has been used. $\square$

Applying P_v to Theorem A.3 removes the local residual exactly. Applying $I - P_v$ retains it. Decomposing each D_i into its projected and normal parts produces the three-state alphabet $\{R, D^P, D^N\}$ used by the mixed-mask dynamic program.

A.3. Universal homogeneous constants.

Theorem A.4 (Universal k and $k - 1$ estimates). *Suppose $\|\mu_v\|_{\text{op}} \leq M$ and $\|r_v\|_{\text{op}} \leq \rho$ at every internal vertex. If T has k internal vertices and reduced leaf-norm product L_T , then*

$$E_T^{\text{amb}} \leq k\rho M^{k-1}L_T, \quad E_T^P = E_T^{\text{red}} \leq (k-1)\rho M^{k-1}L_T, \quad E_T^N \leq k\rho M^{k-1}L_T.$$

Proof. The claims are trivial for a leaf. At an internal vertex v , multilinear telescoping in any fixed slot order gives

$$\begin{aligned} & \|\mu_v(F_1, \dots, F_a) - \mu_v(R_1, \dots, R_a)\| \\ & \leq M \sum_{j=1}^a \|F_j - R_j\| \prod_{i < j} \|R_i\| \prod_{i > j} \|F_i\|. \end{aligned}$$

Assume the ambient estimate for every child. Lemma A.1 shows that the j -th summand is at most $k_{c_j} \rho M^{k_v-1} L_v$. Because $\sum_j k_{c_j} = k_v - 1$, the total propagated-child contribution is at most $(k_v - 1) \rho M^{k_v-1} L_v$. The local residual is bounded by

$$\rho \prod_j \|R_j\| \leq \rho M^{k_v-1} L_v.$$

Their sum proves the ambient estimate. For the projected error,

$$P_v D_v = P_v (\mu_v(F_1, \dots, F_a) - \mu_v(R_1, \dots, R_a)),$$

so the root local residual is absent and only the coefficient $k_v - 1$ remains. Proposition A.2 identifies the reduced error with this projected error. For the normal error, the local residual is retained, and the ambient argument gives coefficient k_v . Induction closes the proof. $\square$

A.4. Optimal telescoping order. Let $e_i, r_i, f_i, G_i \geq 0$, set $w_i = G_i e_i$, and for a permutation π define

$$C(\pi) = \sum_t w_{\pi_t} \prod_{s < t} r_{\pi_s} \prod_{s > t} f_{\pi_s}.$$

Theorem A.5 (Pair-exchange ordering). *Put $d_i = f_i - r_i$. A global minimizer of C is obtained by placing the slots with $d_i > 0$ first, then those with $d_i = 0$, then those with $d_i < 0$, and sorting each nonzero sign class by nondecreasing w_i/d_i . Slots with $w_i = d_i = 0$ are indifferent.*

Proof. Consider adjacent slots i, j . Prefix and suffix products are common to the two possible orders. Thus i before j is no worse precisely when

$$w_i f_j + r_i w_j \leq w_j f_i + r_j w_i \iff w_i d_j \leq w_j d_i.$$

If both d_i, d_j are nonzero and have the same sign, division by the positive product $d_i d_j$ gives $w_i/d_i \leq w_j/d_j$. If $d_i > 0$ and $d_j < 0$, the exchange inequality holds automatically, so every positive-denominator slot precedes every negative one. The cases with one zero denominator place the zero class between them. Repeatedly exchanging inverted adjacent pairs never increases C and terminates at the stated order. Every permutation can be transformed in this way, proving global optimality within the declared telescoping family. $\square$

A.5. Mixed-mask and path-sum certificates.

Theorem A.6 (Sound dynamic program). *Assume certified bounds for every full, projected-output, and normal-output mixed block of each node law. The recursion retaining $B_v^F, B_v^R, B_v^A, B_v^P, B_v^N$ and all non-all- R state vectors in $\{R, D^P, D^N\}^{a_v}$ returns valid upper bounds. Its work is*

$$O(|T|3^{a_{\max}} + |T|a_{\max} \log a_{\max}).$$

Proof. At a leaf all five bounds are exact. At an internal node, expand by Theorem A.3 and then split every child error orthogonally into its D^P and D^N parts. Every resulting term has one of three states in each slot. Multiplying the corresponding certified mixed-block norm by the child magnitude bounds and summing all state vectors except all R is a direct application of the operator-norm inequality. The local residual contributes $\lambda_v = \rho_v \prod_i B_{c_i}^R$ only to full and normal output. Taking the minimum of independently valid upper bounds preserves validity.

There are $3^{a_v} - 1$ nontrivial state vectors at node v , while Theorem A.5 requires sorting a_v slots. Summing these costs over the internal vertices proves the complexity estimate. $\square$

Corollary A.7 (Residual-source path sum). *Fix a valid telescoping order at every ancestor. If $h_{a,j}$ is the certified gain for propagating an error from child slot j through ancestor a , then*

$$B_{T,\text{path}}^{\text{amb}} = \sum_{v \in \text{Int } T} \lambda_v \prod_{(a,j) \in \text{path}(v,\text{root})} h_{a,j}$$

is valid. For projected-root error the root source is omitted and the last gain is a projected-output gain.

Proof. The scalar nodewise recurrence is affine in each local source λ_v . Repeated substitution sends that source through exactly the ancestor edges on its root path and multiplies their gains. Summing over all possible sources gives the formula. Projection annihilates the root normal source, which proves the projected variant. $\square$

A.6. Signed forests and representation perturbations.

Corollary A.8 (Signed-forest triangle certificate). *If $\mathcal{F} = \sum_{\alpha} c_{\alpha} T_{\alpha}$ is a finite compatible signed forest, then the projected discrepancy is bounded by the sum of the per-tree projected certificates weighted by $|c_{\alpha}|$. In particular, the two-term, two-node ternary associator has coefficient at most two.*

Proof. Apply Theorem A.4 to each tree and then the triangle inequality. Each two-node tree contributes $k - 1 = 1$, hence two signed terms contribute at most two. Syntactically identical trees may first be combined exactly. $\square$

Proposition A.9 (Approximation plus projection). *Suppose $\|\mu_v\| \leq M$, $\|\mu_v - \hat{\mu}_v\| \leq \delta$, and $\|\hat{\mu}_v\| \leq M + \delta$. For a k -node homogeneous tree,*

$$E_{\text{repr}} \leq k\delta(M + \delta)^{k-1}L,$$

and a closure coefficient $c \in \{k, k-1\}$ may be split as

$$c\rho(M + \delta)^{k-1}L = c\rho M^{k-1}L + c\rho[(M + \delta)^{k-1} - M^{k-1}]L.$$

Proof. Insert the full ambient $\hat{\mu}$ -tree between the exact ambient tree and the recursively projected approximate tree. Multilinear telescoping over the k node-law replacements bounds each replacement by $\delta(M + \delta)^{k-1}L$, giving the representation term. Apply Theorem A.4 to the approximate law and add and subtract $c\rho M^{k-1}L$; the displayed closure and interaction terms are an algebraic partition of that valid upper bound. No equality with the observed total error is asserted. $\square$

APPENDIX B. EXTREMIZER TABLES

The complete machine-readable extremizer registry stores the mathematical object hash, construction, interval lower, theorem upper, and gap for every block-A and block-B cell. Table 7 is the compact body summary; Figures 7 and 9 expose selected profiles. The supplementary vector atlases under `figures/atlas_fig04_*` retain topology-resolved tiles without compressing them into an unreadable main panel.

APPENDIX C. REPRODUCIBILITY APPENDIX

All calculations were executed inside the single repository. The canonical base pipeline validates the budget, enumerates trees, runs exact and base blocks, aggregates scientific instances, builds generated tables and vector figures, compiles both manuscripts, renders every page, runs adversarial audits, and emits the strict release status. Every run directory includes resolved configuration, command, environment, hardware, source commit, mathematical-object and input hashes, tensors, metrics, certificates, logs, and artifact hashes.

The software companion gives the API and command-level details. The strict gate requires a clean immutable rerun, no unexplained bound violation, all four reviews at least acceptable as a preprint, visual PDF inspection, and an established novelty status. Until those conditions hold, the presence of a compiled PDF is not a release authorization.

TABLE 1. Prior-art overlap matrix generated from the v3 registry. A bounded search cannot establish novelty.

area	overlap	v3 difference	novelty status
Yau, Colored Operads	The v3 type grammar and ordered tree model.	Orthogonal projection defects and quantitative nodewise certificates are additional analytic data.	standard restriction
Loday and Vallette, Algebraic Operads	Signed forests and insertion-tree syntax.	Quantitative normal leakage is not the monograph's central problem.	known-bound specialization
Johnson, Approximately multiplicative maps between Banach algebras	Quantitative stability language for algebraic structure.	V3 propagates nodewise normal projection defects on a fixed finite tree rather than correcting a map globally.	novelty not established
Higham, Accuracy and Stability of Numerical Algorithms	Local error propagation and evaluation-order comparisons.	Exact multilinear projector residuals and typed tree topology.	known-bound specialization
Combettes and Pesquet, Lipschitz Certificates for Layered Network Structures	Dynamic propagation of local operator summaries.	V3 uses multilinear branching, normal/projected blocks, and source residuals.	novelty not established
Gehr et al., AI2	Nodewise abstract summaries and recursively propagated enclosures.	No neural activation or robustness property; exact multilinear projector decomposition.	novelty not established
Hackbusch and Kuehn, A New Scheme for the Tensor Representation	Topology-dependent finite tree calculations.	V3 studies recursive projection of law outputs rather than rank truncation of one tensor.	novelty not established
Ballani and Grasedyck, Tree Adaptive Approximation in the Hierarchical Tensor Format	Tree topology as an experimental factor.	Constants concern closure and recursive projection, not hierarchical ranks.	novelty not established
Kolda and Bader, Tensor Decompositions and Applications	Secondary CP representation budget.	CP error is separated from closure error.	known-bound specialization
Moore, Kearfott, and Cloud, Introduction to Interval Analysis	Small explicit lower-construction enclosures.	Application to tree constant ratios only.	standard multilinear bound
Lasserre, Global Optimization with Polynomials and the Problem of Moments	Proposed small-case global certificates.	V3 does not claim an SDP certificate unless solver and relaxation status are independently validated.	standard multilinear bound
Egger et al., Structure-Preserving Model Reduction	Projection hypotheses and quantified loss of structure.	Static finite multilinear composition trees and closure leakage.	novelty not established

TABLE 2. Typed notation generated from the implementation contract.

symbol	generated meaning
V_τ, W_τ	ambient and reduced Hilbert spaces of type τ
Q_τ, P_τ	isometry and orthogonal projector $P_\tau = Q_\tau Q_\tau^*$
F_v, R_v	ambient and recursively projected values at node v
D_v^P, D_v^N	tangent and normal components of $F_v - R_v$
M_v, ρ_v	local operator-norm and closure-residual bounds
B_v^F, B_v^R	certificates for $\ F_v\ $ and $\ R_v\ $
B_v^A, B_v^P, B_v^N	ambient, projected, and normal error certificates
C_T^{amb}, C_T^P	best universal constants for a typed tree T

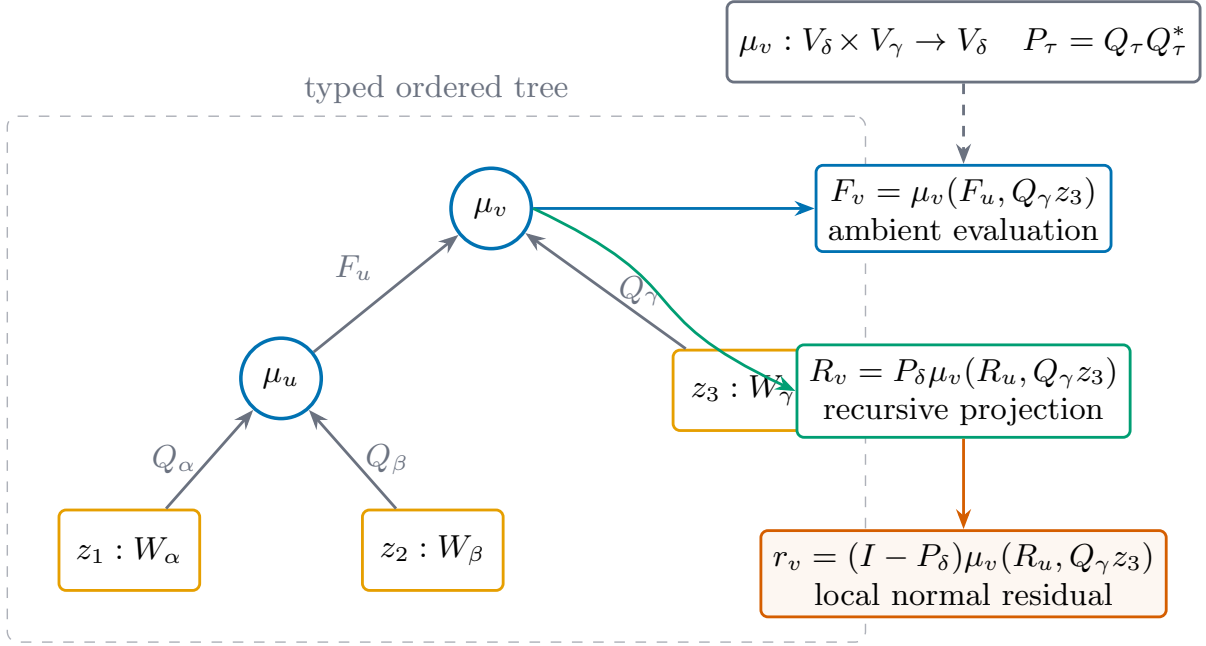


FIGURE 1. Typed ordered-tree semantics. Types determine the ambient spaces, isometries, projectors, and admissible node laws; ambient evaluation, recursive projection, and the local normal residual are distinct operations. This exact definitional diagram has no sampling uncertainty and supports no sharpness or novelty claim.

TABLE 3. Generated nodewise formulas. Each row is an upper-bound contract, not an equality with observed error.

certificate	generated formula	status
homogeneous ambient	$k\rho M^{k-1} \prod_\ell \ z_\ell\ $	proved
homogeneous projected	$(k-1)\rho M^{k-1} \prod_\ell \ z_\ell\ $	proved
nodewise subset	$\rho_v \prod_i B_i^R + M_v \sum_{\emptyset \neq S} \prod_{i \in S} B_i^A \prod_{j \notin S} B_j^R$	proved
path sum	$\sum_v \rho_v \prod_{e \in \pi(v)} L_e$	proved upper
optimized telescoping	$\min_\pi \sum_j w_{\pi_j} (f_{\pi_j} - r_{\pi_j}) \prod_{i < j} f_{\pi_i} \prod_{i > j} r_{\pi_i}$	proved upper

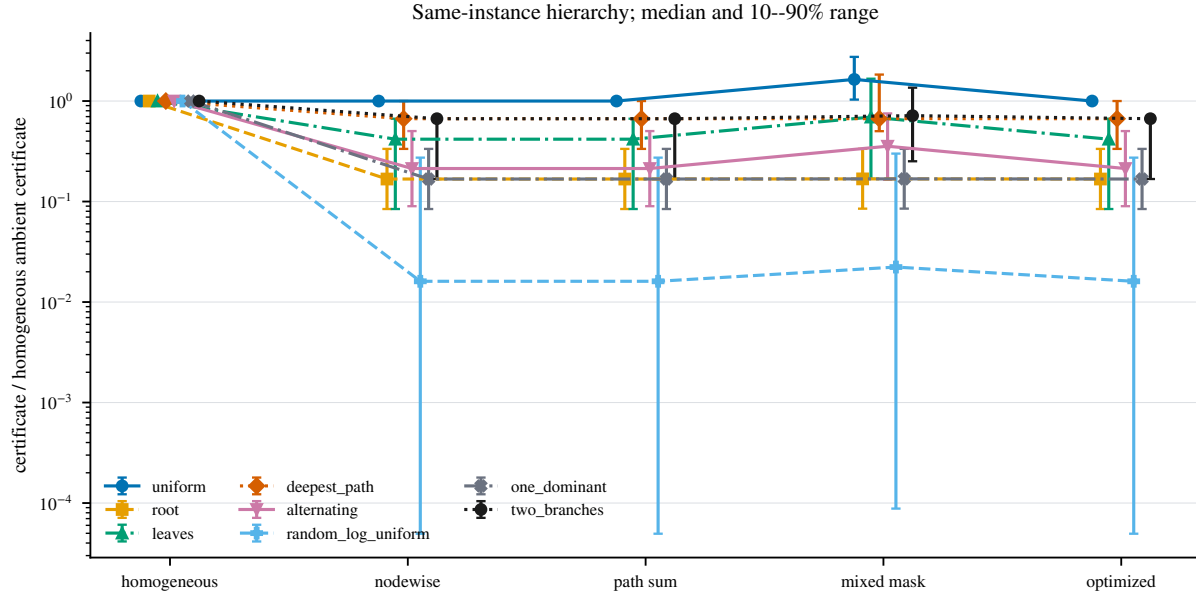


FIGURE 3. Bound hierarchy on block D (4,320 unique registered instances; declared seed count 10, no optimizer restarts). The metric is each sound certificate divided by the homogeneous ambient certificate; points are medians and bars are 10–90 percent instance quantiles, not seed uncertainty. The homogeneous theorem is the reference. The data support instancewise tightening for nonuniform local patterns, but do not prove any certificate globally optimal.

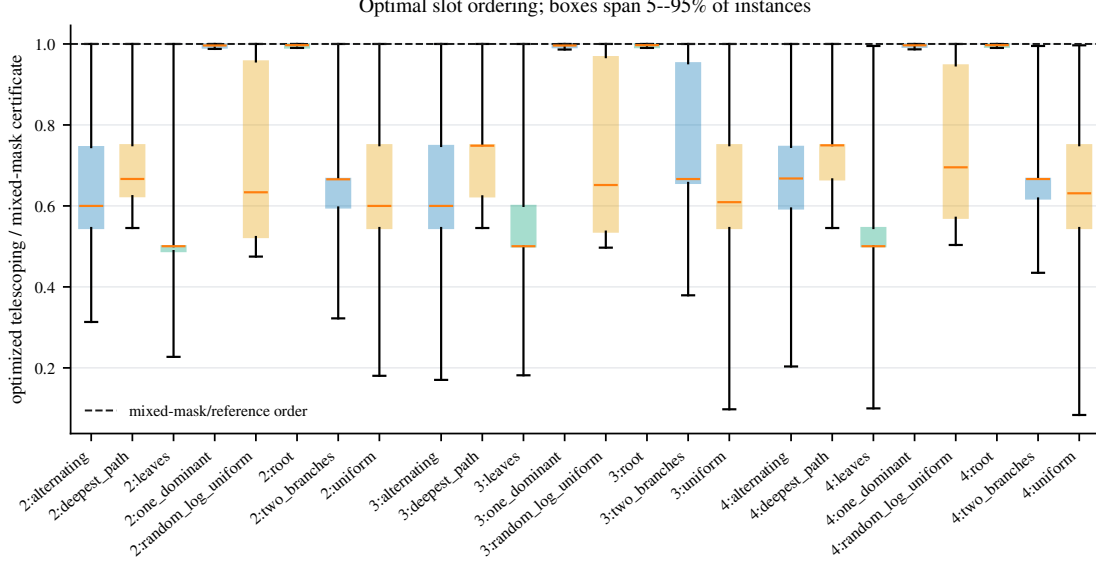


FIGURE 4. Optimal telescoping order on all 4,320 block-D instances (declared seed count 10; no optimizer restart is used by the exact sorting rule). The metric is optimized-order certificate divided by the registered mixed-mask/reference-order certificate; boxes show median, quartiles, and 5–95 percent mathematical-instance variation. The pair-exchange theorem is the reference. The result supports never-worse and sometimes smaller certificates, but not optimality among all possible non-telescoping proof strategies.

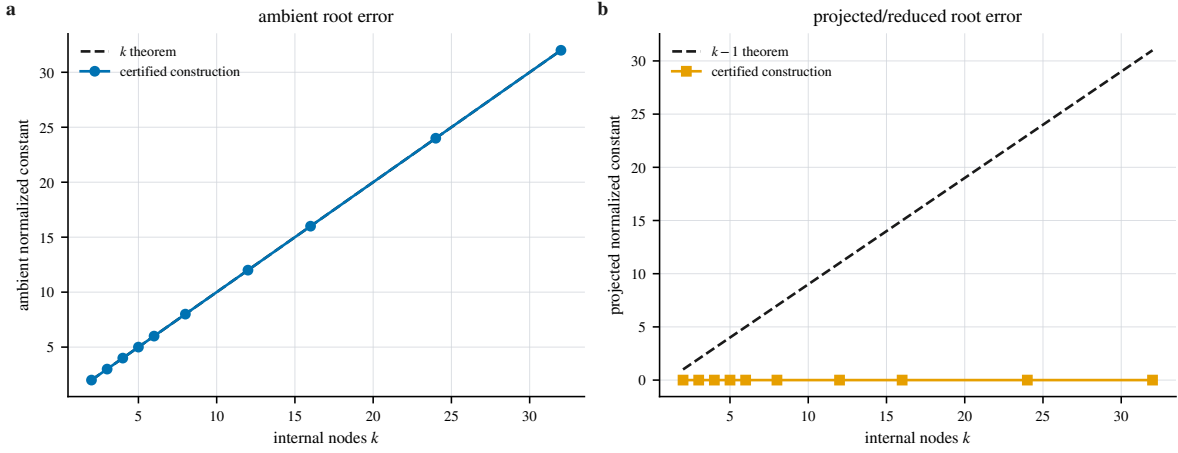


FIGURE 5. The k versus $k - 1$ theorem on 10 registered binary left-comb instances (block B, dimension two, smallest η ; interval-certified lower construction, no seed uncertainty). Ambient and projected errors are normalized by ηM^{k-1} and compared with the proved universal coefficients. The ambient construction matches the linear coefficient in these cells, whereas the displayed small- η projected construction remains far below $k - 1$; projected fixed- η sharpness and novelty remain unresolved.

TABLE 5. Registered homogeneous ambient constants from block B. Lower values are interval-enclosed construction ratios.

arity	k	best ambient lower	theorem upper	lower/upper
2	2	2.000	2.000	1.000
2	3	3.000	3.000	1.000
2	4	4.000	4.000	1.000
2	5	5.000	5.000	1.000
2	6	6.000	6.000	1.000
2	8	8.000	8.000	1.000
2	12	12.000	12.000	1.000
2	16	16.000	16.000	1.000
2	24	24.000	24.000	1.000
2	32	32.000	32.000	1.000
3	2	2.000	2.000	1.000
3	3	3.000	3.000	1.000
3	4	4.000	4.000	1.000
3	5	5.000	5.000	1.000
3	6	6.000	6.000	1.000
3	8	8.000	8.000	1.000
3	12	12.000	12.000	1.000
3	16	16.000	16.000	1.000
3	24	24.000	24.000	1.000
3	32	32.000	32.000	1.000
4	2	2.000	2.000	1.000
4	3	3.000	3.000	1.000
4	4	4.000	4.000	1.000
4	5	5.000	5.000	1.000
4	6	6.000	6.000	1.000
4	8	8.000	8.000	1.000
4	12	12.000	12.000	1.000
4	16	16.000	16.000	1.000
4	24	24.000	24.000	1.000
4	32	32.000	32.000	1.000

TABLE 6. Registered homogeneous projected-root constants. Small- η explicit lower constructions may remain far below the proved upper.

arity	k	best projected-root lower	theorem upper	lower/upper
2	2	1.00×10^{-8}	1.000	1.00×10^{-8}
2	3	3.00×10^{-8}	2.000	1.50×10^{-8}
2	4	6.00×10^{-8}	3.000	2.00×10^{-8}
2	5	1.00×10^{-7}	4.000	2.50×10^{-8}
2	6	1.50×10^{-7}	5.000	3.00×10^{-8}
2	8	2.80×10^{-7}	7.000	4.00×10^{-8}
2	12	6.60×10^{-7}	11.000	6.00×10^{-8}
2	16	1.20×10^{-6}	15.000	8.00×10^{-8}
2	24	2.76×10^{-6}	23.000	1.20×10^{-7}
2	32	4.96×10^{-6}	31.000	1.60×10^{-7}
3	2	1.00×10^{-8}	1.000	1.00×10^{-8}
3	3	3.00×10^{-8}	2.000	1.50×10^{-8}
3	4	6.00×10^{-8}	3.000	2.00×10^{-8}
3	5	1.00×10^{-7}	4.000	2.50×10^{-8}
3	6	1.50×10^{-7}	5.000	3.00×10^{-8}
3	8	2.80×10^{-7}	7.000	4.00×10^{-8}
3	12	6.60×10^{-7}	11.000	6.00×10^{-8}
3	16	1.20×10^{-6}	15.000	8.00×10^{-8}
3	24	2.76×10^{-6}	23.000	1.20×10^{-7}
3	32	4.96×10^{-6}	31.000	1.60×10^{-7}
4	2	1.00×10^{-8}	1.000	1.00×10^{-8}
4	3	3.00×10^{-8}	2.000	1.50×10^{-8}
4	4	6.00×10^{-8}	3.000	2.00×10^{-8}
4	5	1.00×10^{-7}	4.000	2.50×10^{-8}
4	6	1.50×10^{-7}	5.000	3.00×10^{-8}
4	8	2.80×10^{-7}	7.000	4.00×10^{-8}
4	12	6.60×10^{-7}	11.000	6.00×10^{-8}
4	16	1.20×10^{-6}	15.000	8.00×10^{-8}
4	24	2.76×10^{-6}	23.000	1.20×10^{-7}
4	32	4.96×10^{-6}	31.000	1.60×10^{-7}

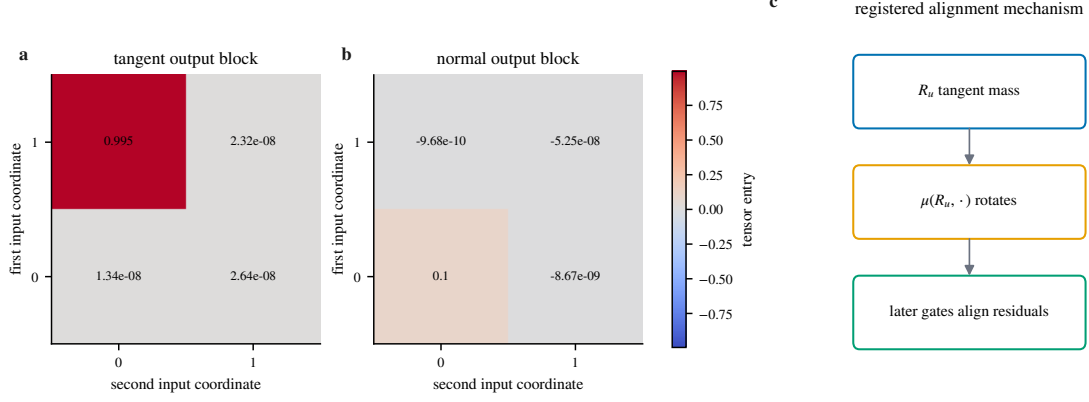


FIGURE 6. Extremizer geometry from the registered v3 smoke run (one explicit rank-one gated-rotation tensor, four gradient trajectories and one derivative-free recheck). Panels (a,b) show the actual tangent and normal output tensor blocks; panel (c) summarizes the certified mechanism. Entries are exact stored inputs up to float64 representation, without statistical uncertainty. The figure supports admissibility and alignment of this lower construction, not uniqueness or global optimality.

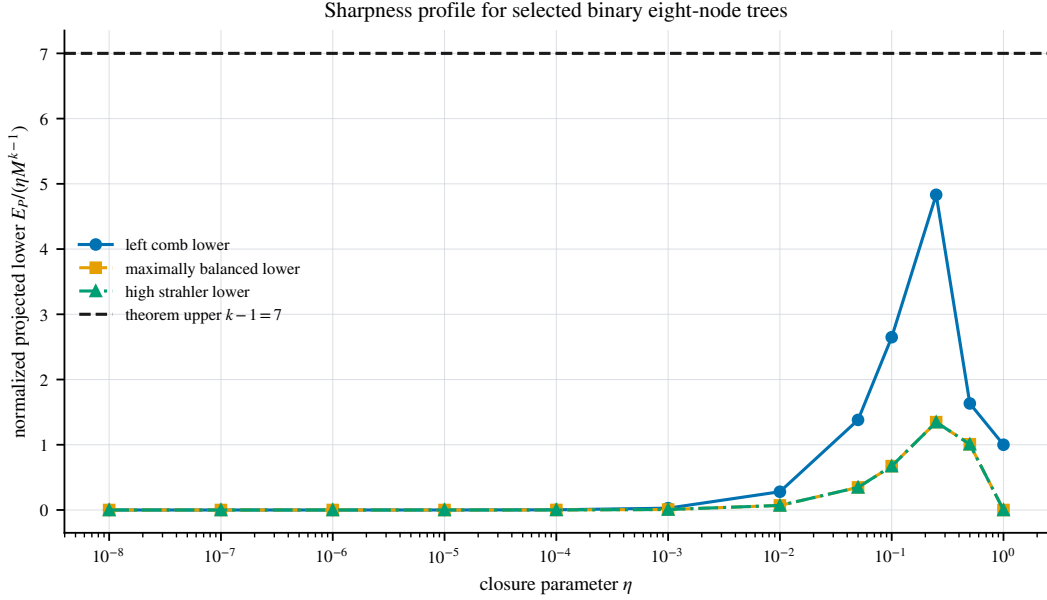


FIGURE 7. Sharpness profile from block B (96 unique binary eight-node instances; the requested 20 seeds and eight restarts remain extended). The metric is the interval-certified explicit projected error divided by ηM^{k-1} ; the dashed reference is the proved $k-1=7$ upper. There is no statistical uncertainty in the explicit construction. The plot supports an admissible lower profile, not a globally optimal $C_T(\eta)$.

TABLE 7. Best registered explicit construction cells by error type.

error	construction	arity	k	η	lower/upper	status
ambient	gated planar rotation	2	1	0.500	1.000	NEAR_OPTIMAL_WITH_CERTIFIED_GAP
normal	gated planar rotation	2	1	0.500	1.000	NEAR_OPTIMAL_WITH_CERTIFIED_GAP
projected	gated planar rotation	2	2	1.000	1.000	NEAR_OPTIMAL_WITH_CERTIFIED_GAP

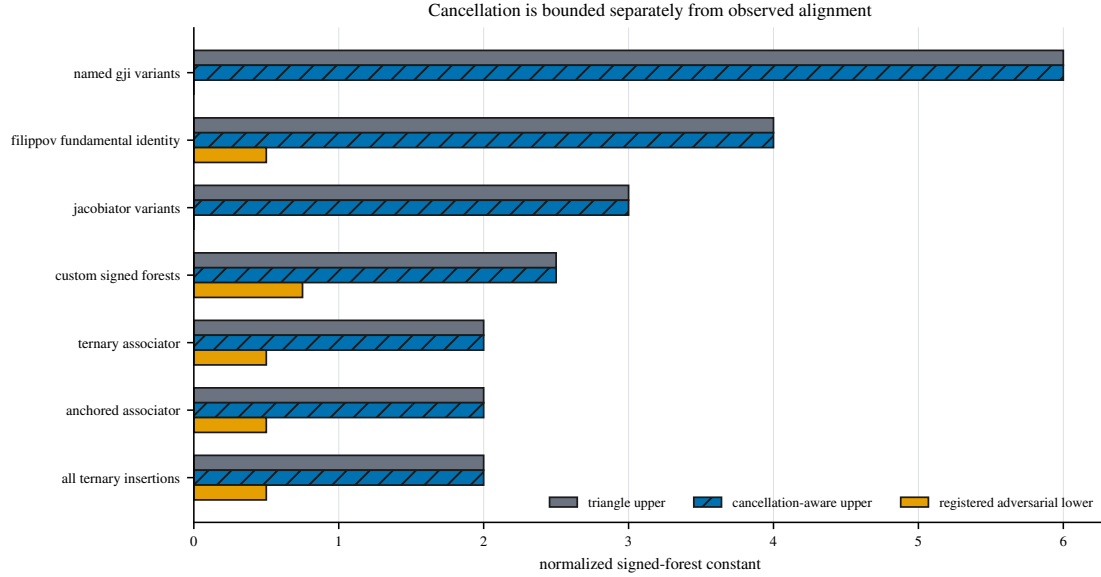


FIGURE 8. Signed-polynomial cancellation across 7 registered forest families and 28 block-G instances (20 optimizer seeds requested; the full restart grid is pending). The exact metric is the normalized signed-forest residual: triangle and syntactic-cancellation bars are certified uppers, while the orange bar is the largest registered explicit/adversarial lower over η . The forest triangle theorem is the reference. The comparison supports improved bookkeeping in declared expressions, not globally sharp associator, FI, or GJI constants.

TABLE 8. Associator and signed-expression constants generated from block G.

expression	triangle	cancellation upper	observed	observed/triangle	search status
all_ternary_insertions	2.000	2.000	0.500	0.250	EXTENDED _PENDING _RESOURCE_GATE
anchored_associator	2.000	2.000	0.500	0.250	EXTENDED _PENDING _RESOURCE_GATE
custom_signed_forests	2.500	2.500	0.750	0.300	EXTENDED _PENDING _RESOURCE_GATE
filippov_fundamental_identity	4.000	4.000	0.500	0.125	EXTENDED _PENDING _RESOURCE_GATE
five_input_ternary_associator	2.000	2.000	0.500	0.250	EXTENDED _PENDING _RESOURCE_GATE
jacobiator_variants	3.000	3.000	0	0	EXTENDED _PENDING _RESOURCE_GATE
named_gji_variants	6.000	6.000	2.22×10^{-12}	3.70×10^{-13}	EXTENDED _PENDING _RESOURCE_GATE

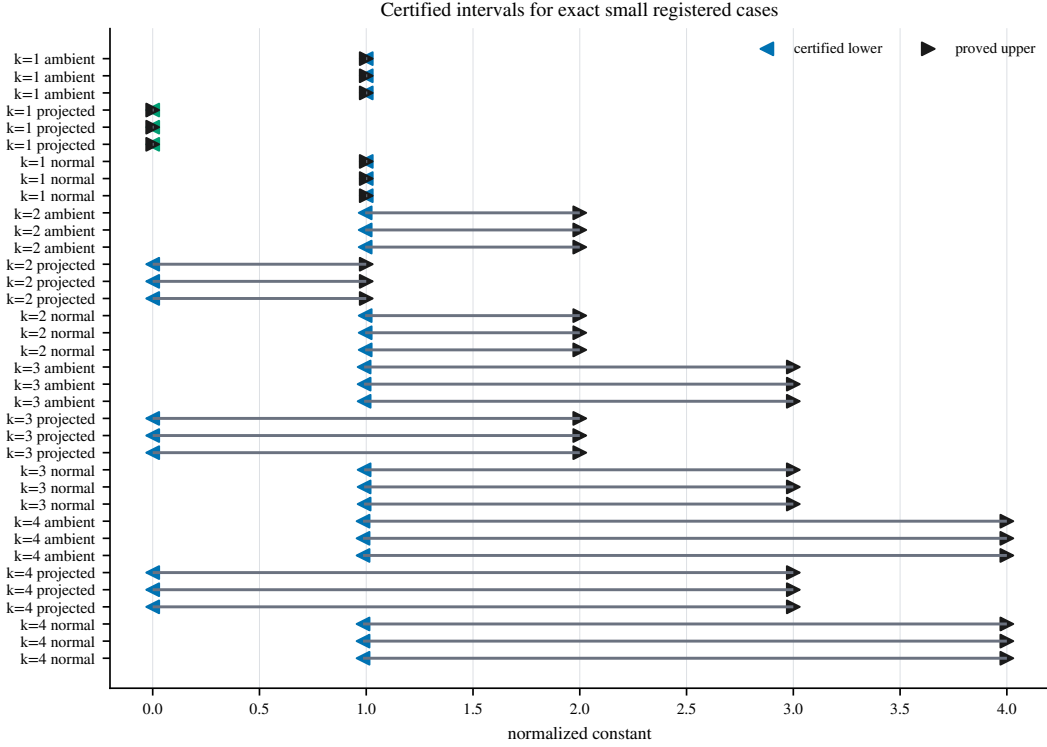


FIGURE 9. Certified optimality gaps for 36 block-A binary small cases at $\eta = 0.1$ (no random seeds or restarts). Horizontal intervals join the directed-interval explicit lower to the proved upper; green lower marks the 3 globally certified zero/small cases. The homogeneous k and $k - 1$ theorems are the references. Closed intervals establish only the marked exact cases; open intervals are reported as unresolved, not as numerical optima.

TABLE 9. Exact and certified small cases selected from block A.

arity	k	error	lower	upper	global certificate
2	1	ambient	1.000	1.000	yes
2	1	normal	1.000	1.000	yes
2	1	projected	0	0	yes
2	2	ambient	1.000	2.000	no
2	2	normal	1.000	2.000	no
2	2	projected	0	1.000	no
2	3	ambient	1.000	3.000	no
2	3	normal	1.000	3.000	no
2	3	projected	0	2.000	no
2	4	ambient	1.000	4.000	no
2	4	normal	1.000	4.000	no
2	4	projected	0	3.000	no
3	1	ambient	1.000	1.000	yes
3	1	normal	1.000	1.000	yes
3	1	projected	0	0	yes
3	2	ambient	1.000	2.000	no
3	2	normal	1.000	2.000	no
3	2	projected	0	1.000	no
3	3	ambient	1.000	3.000	no
3	3	normal	1.000	3.000	no
3	3	projected	0	2.000	no
3	4	ambient	1.000	4.000	no
3	4	normal	1.000	4.000	no
3	4	projected	0	3.000	no

TABLE 10. Optimality-gap registry aggregated by named error.

error	cells	exact	near	median abs. gap	max rel. gap
ambient	4275	0	258	3.000	1.000
normal	1395	0	30	2.701	1.000
projected	4275	30	21	3.000	1.000

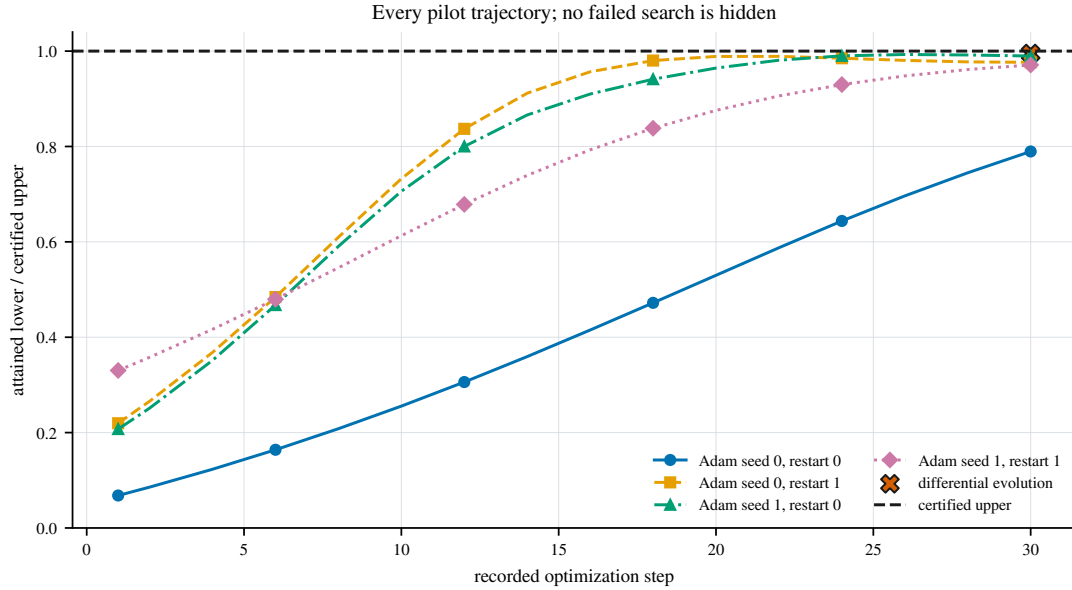


FIGURE 10. Adversarial optimization convergence for the registered smoke calibration (5 displayed trajectories: four Adam starts and one independent differential-evolution recheck; one mathematical instance). The metric is attained normalized lower divided by the certified upper; every recorded step and all searches are shown, with no confidence interval inferred from restarts. The upper certificate is the reference. Agreement near one supports a strong empirical lower for this instance, not a global optimizer certificate or completion of the 460,800-trajectory extended grid.

TABLE 11. Exhaustive ordered-tree enumeration statistics. Cross-grammar hash reuse is intentional.

grammar	occurrences	unique hashes	internal nodes	max depth
binary	2055	2055	1–8	8
mixed_2_3_4	78879	78879	1–5	5
quaternary	167	167	1–4	4
ternary	344	344	1–5	5

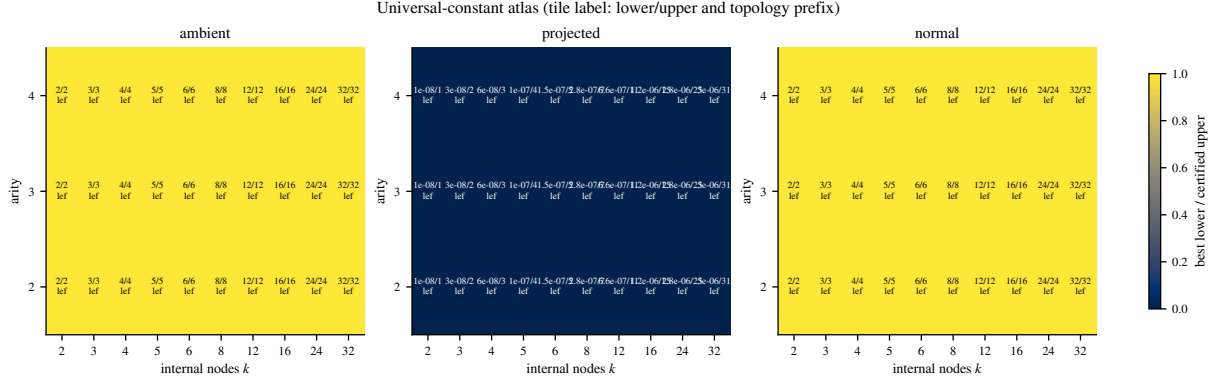


FIGURE 11. Universal constant atlas from block B (240 dimension-two instances selected from 2,880; 20 seeds and eight restarts are specified for the pending extended search). Tile color is the best certified explicit lower divided by the homogeneous upper; text gives lower/upper and the maximizing topology prefix. Interval enclosure is used for the construction, with no sampling uncertainty. The atlas supports exact or near-matching cells where the ratio is one, but does not establish fixed- η optimality across the full matrix.

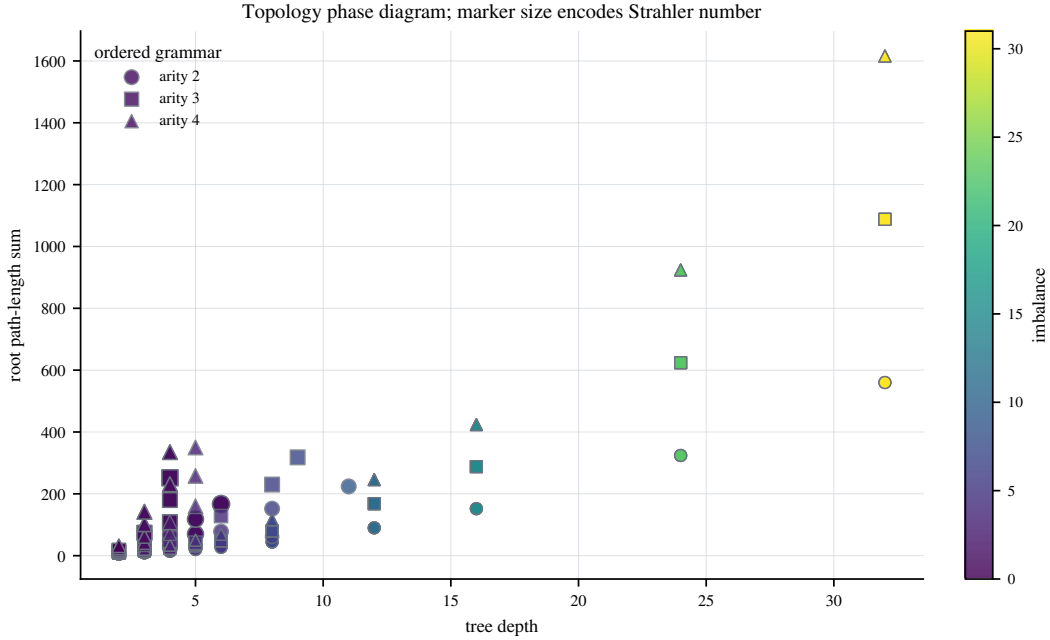


FIGURE 12. Topology phase diagram from block B (240 unique $\eta = 0.1$ instances; requested optimizer seeds/restarts are pending). Depth and root path-length sum define the axes, color is imbalance, size is Strahler number, and a vermilion edge marks an explicit projected lower/upper ratio above 0.9. These are exact topology statistics with interval lower values. The plot supports topology-dependent certificate behavior, not a causal phase transition or universal extremal topology.

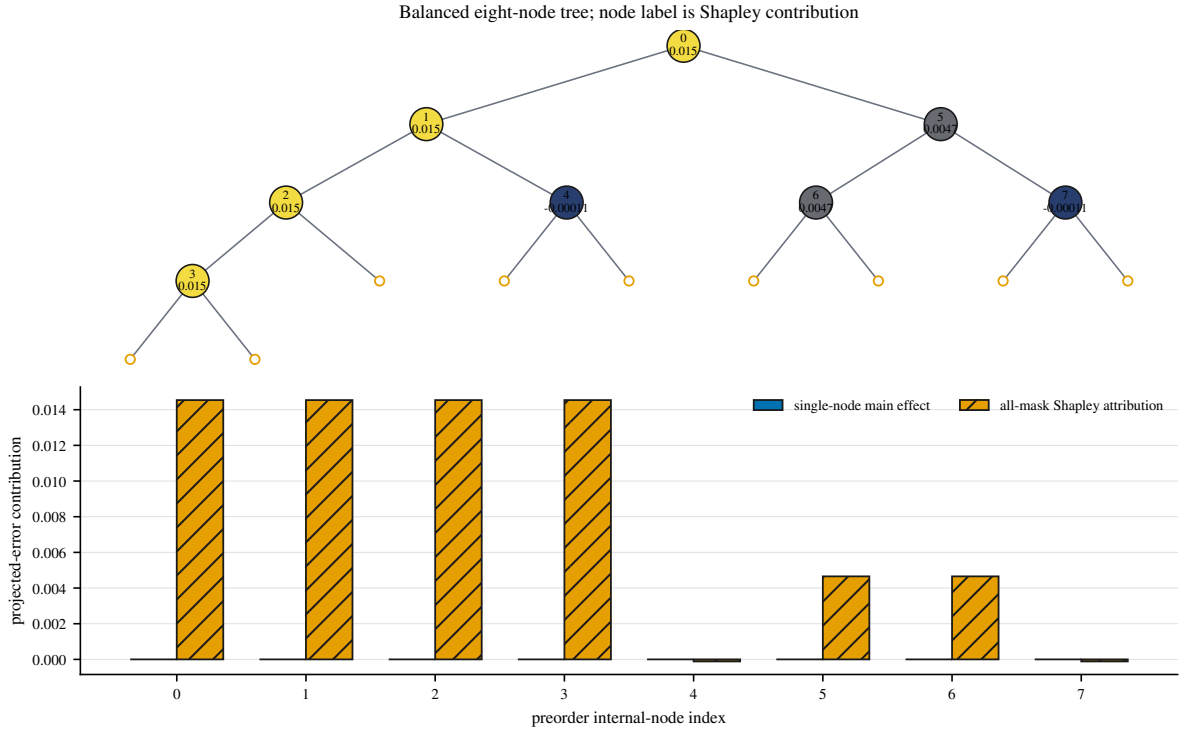


FIGURE 13. Node contribution map for one balanced eight-node tree from block F (eight internal nodes, all $2^8 = 256$ leakage masks, no seed or restart sampling). The metric is projected root error: the tree labels show Shapley attribution and the lower panel compares single-node main effects with all-mask Shapley values; numerical efficiency residual is below 10^{-12} . The exhaustive mask calculation supports attribution on this registered construction, not a unique physical causal interpretation.

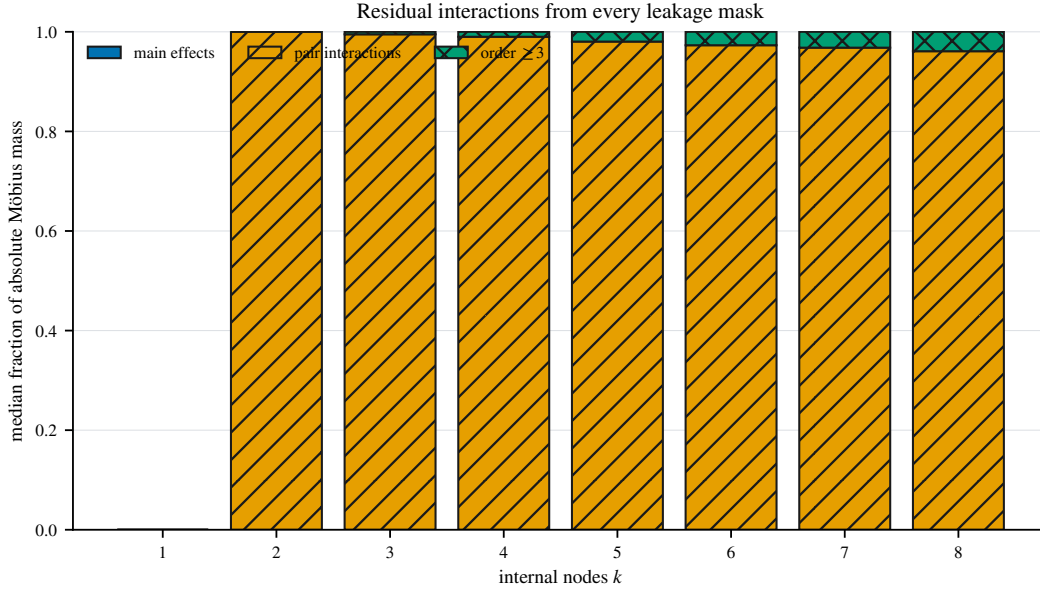


FIGURE 14. Residual interaction decomposition from all 1,530 block-F mask evaluations (24 trees, no seeds or optimizer restarts). For each tree, the exact set function of projected error is Möbius-inverted; bars show topology-median fractions of absolute coefficient mass from first-, second-, and higher-order subsets. The reference is the exact subset expansion. This supports nonadditive local-error interactions in the explicit construction, not sign-free additivity or universality.

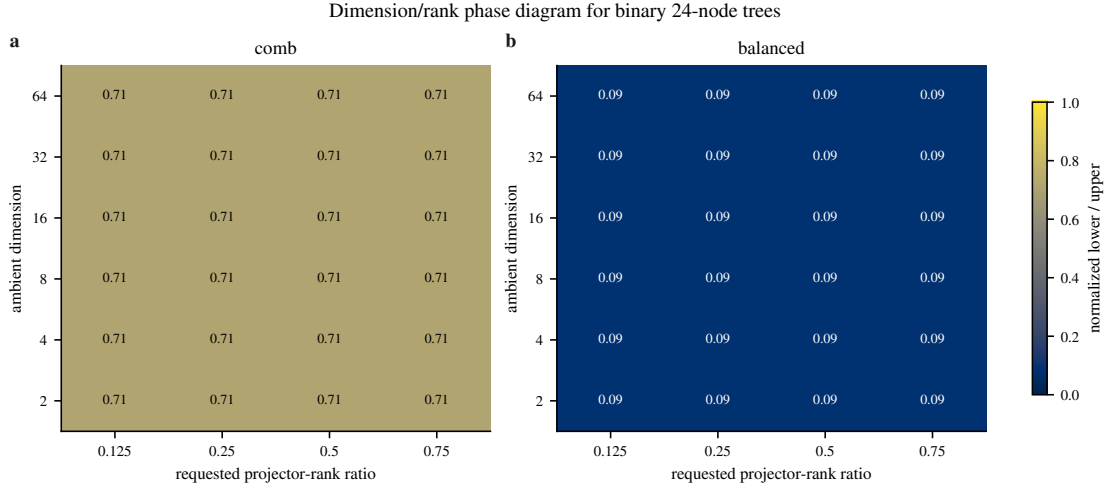


FIGURE 15. Dimension/rank phase diagram from 72 block-C instances (binary 24-node trees, $\eta = 0.1$; ten optimizer seeds requested but not executed in the analytic base). Each tile is the embedded dimension-two projected lower divided by the $k-1$ upper; values are interval-enclosed and have no seed uncertainty. The embedding theorem is the reference. The grid supports dimension-independent admissibility of the construction, but no certified improvement from extra dimensions or ranks.

TABLE 12. Dimension/rank interaction. The registered lower is an embedded dimension-two construction.

dimension	rank ratio	rank	best normalized lower	upper	improvement certified
2	0.125	1	16.265	2.000	no
2	0.250	1	16.265	2.000	no
2	0.500	1	16.265	2.000	no
2	0.750	1	16.265	2.000	no
4	0.125	1	16.265	2.000	no
4	0.250	1	16.265	2.000	no
4	0.500	2	16.265	2.000	no
4	0.750	3	16.265	2.000	no
8	0.125	1	16.265	2.000	no
8	0.250	2	16.265	2.000	no
8	0.500	4	16.265	2.000	no
8	0.750	6	16.265	2.000	no
16	0.125	2	16.265	2.000	no
16	0.250	4	16.265	2.000	no
16	0.500	8	16.265	2.000	no
16	0.750	12	16.265	2.000	no
32	0.125	4	16.265	2.000	no
32	0.250	8	16.265	2.000	no
32	0.500	16	16.265	2.000	no
32	0.750	24	16.265	2.000	no
64	0.125	8	16.265	2.000	no
64	0.250	16	16.265	2.000	no
64	0.500	32	16.265	2.000	no
64	0.750	48	16.265	2.000	no

TABLE 13. Precision and conditioning controls.

precision	case	max residual	backward error	condition est.	min bound margin
arbitrary_precision	ill_conditioned	0	0	1.00×10^8	3.00×10^{-8}
arbitrary_precision	near_bound	0	0	1,000.000	0.003
arbitrary_precision	optimizer_stability	2.22×10^{-16}	2.09×10^{-16}	2.000	0.438
arbitrary_precision	small_exact	0	0	10.000	0.241
complex128	ill_conditioned	0	0	1.00×10^8	3.00×10^{-8}
complex128	near_bound	0	0	1,000.000	0.003
complex128	optimizer_stability	0	0	2.000	0.438
complex128	small_exact	0	0	10.000	0.241
complex64	ill_conditioned	0	0	1.00×10^8	3.00×10^{-8}
complex64	near_bound	8.48×10^{-13}	8.48×10^{-13}	1,000.000	0.003
complex64	optimizer_stability	4.04×10^{-8}	3.80×10^{-8}	2.000	0.438
complex64	small_exact	6.99×10^{-10}	6.99×10^{-10}	10.000	0.241
exact_rational	ill_conditioned	0	0	1.667	0.547
exact_rational	near_bound	0	0	1.667	0.547
exact_rational	optimizer_stability	0	0	1.667	0.547
exact_rational	small_exact	0	0	1.667	0.547
float32	ill_conditioned	0	0	1.00×10^8	3.00×10^{-8}
float32	near_bound	8.48×10^{-13}	8.48×10^{-13}	1,000.000	0.003
float32	optimizer_stability	4.04×10^{-8}	3.80×10^{-8}	2.000	0.438
float32	small_exact	6.99×10^{-10}	6.99×10^{-10}	10.000	0.241
float64	ill_conditioned	0	0	1.00×10^8	3.00×10^{-8}
float64	near_bound	0	0	1,000.000	0.003
float64	optimizer_stability	0	0	2.000	0.438
float64	small_exact	0	0	10.000	0.241

TABLE 14. CPU/GPU parity. The tolerance is a validation threshold, not a theorem error.

precision	cases	max abs. difference	median difference	result
complex128	4	0	0	pass
complex64	4	1.92×10^{-8}	5.60×10^{-9}	pass
float32	4	1.92×10^{-8}	5.60×10^{-9}	pass
float64	4	0	0	pass

TABLE 15. Failure and counterexample registry. Open scientific questions are retained as blockers.

item	count	disposition
registered run failures	0	none
invalid typed edges	48	rejected before evaluation
zero projected one-node constant	30	exact control
fixed- η universal sharpness	1	open; no optimality claim
extended optimizer grid	1	resource-gated and pending

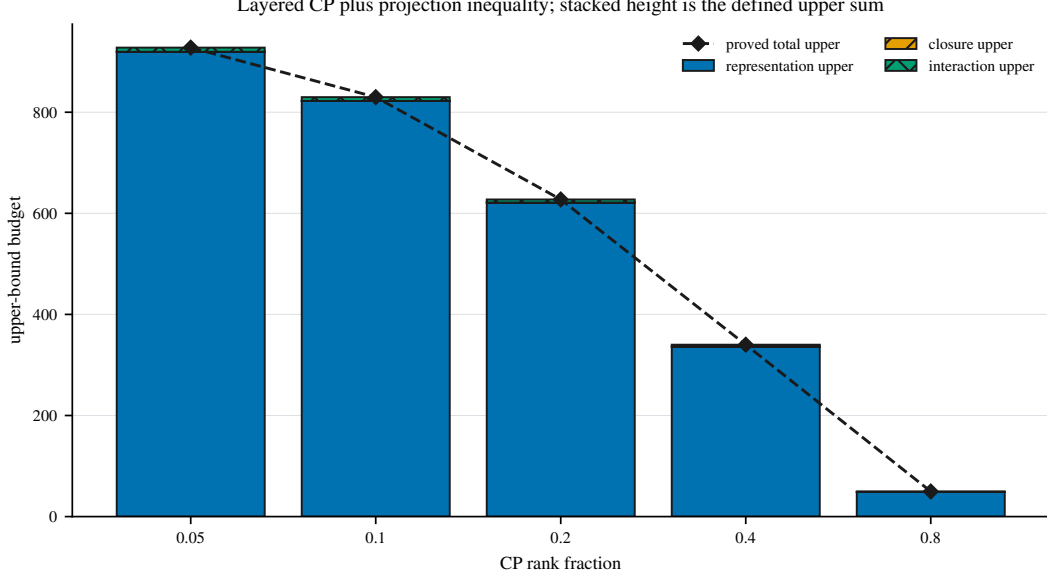


FIGURE 16. CP plus projection budget for 5 selected block-H instances (dimension 64, eight-node balanced tree, $\eta = 0.01$; ten seeds are declared for the extended empirical matrix). Bars are the separately proved representation, closure, and interaction upper terms; their stack is the definition of the theorem upper and does not assert equality with error. The layered CP/projection inequality is the reference. The figure supports budget separation, not tightness or CP identifiability.

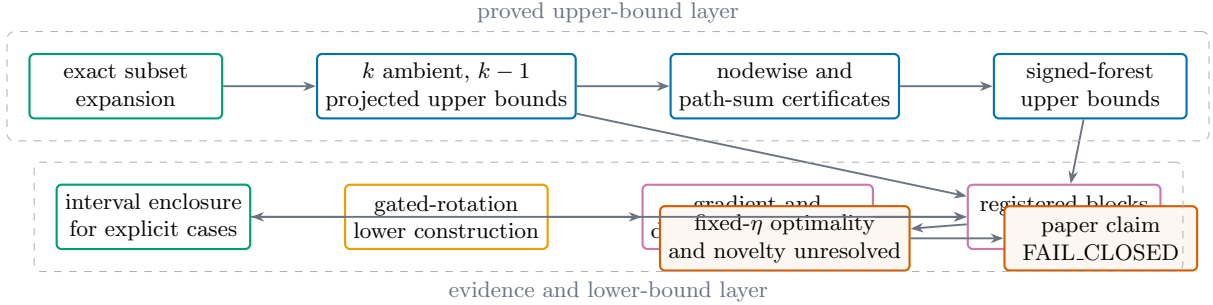


FIGURE 17. Claim/evidence/extremizer dependency graph generated from the v3 theorem and evidence registries. Proved upper bounds, certified or empirical lower evidence, experiments, and unresolved novelty are separate epistemic layers. The graph supports traceability only; it does not turn a search result into a theorem or authorize release.

TABLE 16. Generated CP plus projection budget. Components sum to a theorem upper, not to observed error.

dim.	rank frac.	rank	tensor err.	representation	closure	interaction	total upper
16	0.050	1	0.968	457.967	0.050	4.070	462.087
16	0.100	2	0.935	394.174	0.050	3.619	397.843
16	0.200	3	0.901	336.334	0.050	3.198	339.582
16	0.400	6	0.791	195.688	0.050	2.102	197.840
16	0.800	13	0.433	24.042	0.050	0.428	24.521
32	0.050	2	0.968	457.967	0.050	4.070	462.087
32	0.100	3	0.952	425.309	0.050	3.841	429.199
32	0.200	6	0.901	336.334	0.050	3.198	339.582
32	0.400	13	0.771	176.675	0.050	1.942	178.667
32	0.800	26	0.433	24.042	0.050	0.428	24.521
64	0.050	3	0.976	474.879	0.050	4.187	479.116
64	0.100	6	0.952	425.309	0.050	3.841	429.199
64	0.200	13	0.893	322.770	0.050	3.097	325.917
64	0.400	26	0.771	176.675	0.050	1.942	178.667
64	0.800	51	0.451	27.128	0.050	0.469	27.648
128	0.050	6	0.976	474.879	0.050	4.187	479.116
128	0.100	13	0.948	417.384	0.050	3.785	421.218
128	0.200	26	0.893	322.770	0.050	3.097	325.917
128	0.400	51	0.776	181.317	0.050	1.982	183.348
128	0.800	102	0.451	27.128	0.050	0.469	27.648

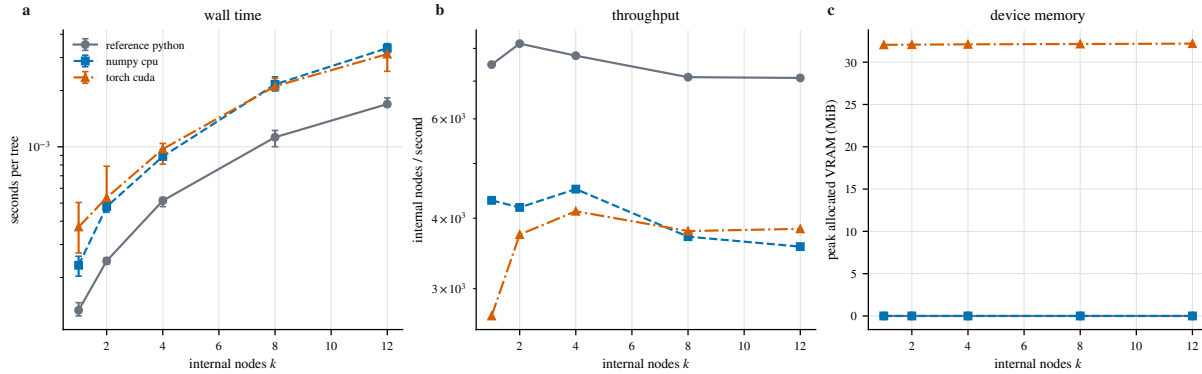


FIGURE 18. Environment-specific scaling benchmark (15 registered backend/size cells; five timing repeats and 15–30 calls per repeat). Panels show median wall time with min–max repeat range, derived node throughput, and peak allocated VRAM for the coordinate reference, vectorized NumPy CPU, and CUDA evaluator; float64 outputs agree exactly in this invariant control. No asymptotic complexity or cross-hardware performance claim is supported.

TABLE 17. Generated reproducibility manifest. The displayed source commit is the development checkpoint and is replaced by the immutable v3 source commit for the canonical final rerun.

field	generated value
source commit	b718f4e51785
scientific instances A–I	15493
enumerated tree occurrences	81445
unique tree hashes	80870
leakage masks	1530
CPU/GPU max difference	1.922e-08
optimizer trajectories requested	460800
optimizer status	EXTENDED_PENDING_RESOURCE_GATE
release status	FAIL_CLOSED_NOVELTY

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